

Healthcare

With nanotechnology, what's beneath our skin is going to be more accessible to us than ever before:

- ▶ Hospitals will benefit greatly from nanotechnology with faster, cheaper diagnostic equipment. The lab-on-a-chip will be able to analyse a patient's ailments in an instant, providing point-of-care testing and drug application.

New contrast agents will float through the bloodstream, lighting up problems such as tumours with incredible accuracy. Diagnostic tests will become more portable, providing time-sensitive diagnostics out in the field in ambulances. New-born children will



The future is Nanomedicine

have their DNA quickly mapped, pointing out future potential problems, allowing us to curtail diseases before they happen.

- ▶ Nanotechnology will aid in the delivery of just the right amount of medicine to the exact spots of the body that need it most. Nanoshells, approximately 100nm in diameter, will float through the body, attaching only to cancer cells. When excited by a laser beam, the nanoshells will give off heat – in effect, cooking the tumour and destroying it. Nanotechnology will create biocompatible joint replacements and artery stents that will last the life of the patient instead of it having to be replaced every few years.

Resources

We undoubtedly are facing a future with curtailed and vanishing resources – be it water or energy. Nanotechnology can enable us to replenish these resources. It can provide new methods to effectively utilise our current energy resources while also presenting new alternatives. Cars will have lighter and stronger engine blocks and frames and will use new additives making fuel more efficient. House lighting will use quantum dots – nanocrystals 5nm across – in order to transform electricity into light instead of wasting away into heat.

- ▶ Nanotechnology will cut costs both, of the solar cells and the equipment